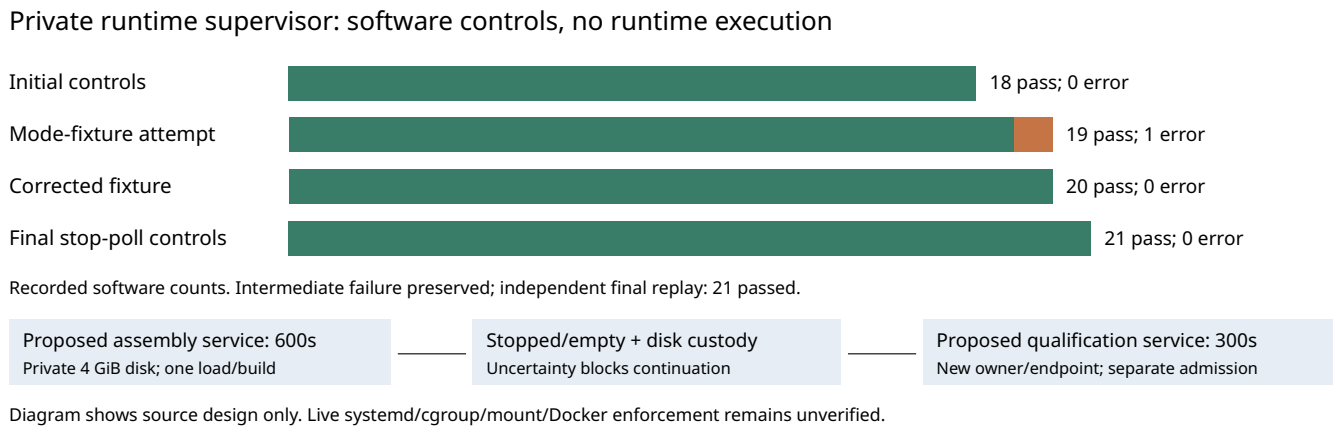
Private runtime supervisor: source and software controls

Implemented a prospective two-phase supervisor with separate assembly and qualification admissions. The final source passed 21 mocked and local temporary-file controls; a separate reviewer inspected the exact source and replayed the 21 controls. No systemd service, daemon, mount, loop device, image load/build, container or cloud job was started by this packet. Both filled binding drafts remain unadmitted.



The proposed assembly unit has 600 total seconds and 60 seconds reserved; qualification has 300 and 30. Each uses 200% CPU quota, 2 GiB memory, zero swap, 256 tasks and an owned 4 GiB ext4 backing file. These are source-defined limits, not observed live enforcement. A service-owned private dockerd starts its own containerd, with private socket/data/exec roots, cgroupfs descendants, network and mount namespaces. Systemd properties and live cgroup controller values are checked before bootstrap; stop escalation fits inside each phase ceiling.

Assembly checks the retained base archive SHA 43a645ed... and 549,662,208 bytes, exact base image 347667d6..., copied context bytes/modes and the COPY-only recipe, then permits one load and one build. Qualification is a distinct unit that locks and verifies the exact stopped predecessor disk. It snapshots four qualifier source files and the exact binding, passes a fresh endpoint receipt to the corrected qualifier-v2, and preserves the callback's held-release containment checks. Private daemon and managed-child identities, source/image joins and endpoint expiry are explicit. JSON admission fields do not authenticate themselves; actual approval and root operator launch are required.

Lifecycle: successful worker exit is retained with RemainAfterExit while invocation and native exit properties are captured. The launcher then stops the owned unit and checks empty cgroup, detached loop device and unchanged disk custody. Stop polling requires both inactive/collected unit and empty cgroup. Unknown or failed states preserve artifacts and block phase continuation. A stopped endpoint is never reported as live. There is no automatic retry, cleanup deletion, signing-key provisioning or permission expansion.

Validation history: the first 18 controls passed. The next 20-control process had 19 passes and one error because the test's declared 0755 input directory inherited the host umask. Its source, outputs and failure remain preserved. The fixture now explicitly sets its declared mode; the same implementation passed all 20. A final bounded-stop polling correction then passed21 controls; the prior20-pass source and receipt are preserved. Controls cover admission/source joins, private command endpoints, finite phase properties, non-main-thread managed-child discovery, readiness-only retry, early failure receipts, invocation identity, deadline rejection, uncertain cgroup state, exact directory modes under 0077 and the four-file qualifier snapshot. All external command methods were replaced with rejecting mocks.

Independent review: assembler reviewer supplied design advice and then reviewed supervisor SHA 2af9bb631a0d73f8118655857a7bb74b4f688fa26628054251196412a85c0256 and test SHA 29cd83febb04022e1eaddd10ca33ea8b7fac6fdf53d1362c23773e16277fa21f. Its 21-control replay passed; no required source corrections remained in that pass. This was not blind design, runtime qualification or execution authorization. Actual systemd, loop/mount, cgroup and private-daemon behavior remains the concrete next validation dependency.

Primary references checked: local systemd.service(5), systemd.kill(5), systemd.resource-control(5), systemd.exec(5), dockerd(8); retained runtime-assembler-v2 and replacement-contract source. Relevant sections cover Type/RuntimeMaxSec/RemainAfterExit, KillMode, resource ceilings/delegation, namespaces and private daemon roots/cgroup parent. Exact commands, source hashes, software receipts and unadmitted inputs are included. Existing matched-cohort planning limits remain unchanged.